

Q-AlloSolve

A Hybrid AI–Quantum Framework for Allosteric Site Prediction and Protein Signal Propagation

A Scalable Framework for Residue Network Analysis, AI-Augmented Biological Inference, Quantum Graph Propagation, and Distributed Quantum Simulation using Fujitsu Quantum Simulator

Final Report Submission

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Submission Date: May 27, 2026

Project Category: Computational Biology / Hybrid AI–Quantum Optimization

Keywords: Allosteric Site Prediction, Protein Signal Propagation, Residue Network Analysis, Quantum Graph Propagation, QUBO, QAOA, AI-Guided Discovery, mpiQulacs, Fujitsu Quantum Simulator, HPC

Submitted to:

Fujitsu Quantum Simulator Challenge Committee

Quantum Simulator Challenge 2026

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Executive Summary

Q-AlloSolve: Hybrid AI–Quantum Framework for Scalable Allosteric Residue Discovery

Problem. Allosteric residue discovery requires identifying distal regulatory residues that control protein function through non-local communication pathways. Classical graph methods often miss weak long-range signals, while quantum-only methods are costly and less effective alone.

Solution Overview.

Component	Role
Protein Graph Modeling	Builds residue communication networks
AI Candidate Filtering	Reduces biological search space
Quantum Propagation	Captures non-local residue influence
QUBO / QAOA	Optimizes residue subset selection
Hybrid Fusion	Combines AI, quantum, and graph evidence
Fujitsu QARP / mpiQulacs	Scaling and distributed simulation evidence

Pipeline.

Protein → Graph → AI Filter → Quantum Propagation → QUBO/QAOA → Rank

Benchmark.

Benchmark Item	Result
Protein systems	KRAS; BCR–ABL1; cardiac myosin; c-Myc
Strict validation targets	KRAS; BCR–ABL1; cardiac myosin
Exploratory target	c-Myc
Largest graph	2528 residues
Weighted graph edges	11,439

Key Results.

Metric	Result
Classical baseline	0.274
AI-only	0.317
Quantum-only	0.413
Hybrid AI–quantum	0.952
Mean best-hit rank	2.62
AI search-space reduction	90–99%+
Hard-case recovery	Classical 0.0 vs Hybrid 1.0
First validated hit	Rank 1 in all strict targets

Residue-Level Validation.

QARP / Scaling Evidence.

Advantages. High recovery accuracy (0.952 top-5 success), strong biological validation, 90–99%+ AI search-space reduction, hard-case recovery, non-local quantum refinement, and Fujitsu-oriented distributed scaling.

Protein	Recovery	Recovered Residues
KRAS	3/4	13, 59, 61
BCR-ABL1	4/4	271, 315, 317, 318
Cardiac myosin	3/4	178, 179, 180

Scaling Metric	Result
Quantum scale	~ 28 qubits
MPI ranks	Up to 8 ranks
Distributed speedup	$\sim 6.96\times$
FusedSWAP reduction	25–37%
Confirmed execution evidence	<code>multi-cpu mpiQulacs logs</code>

Scientific Insight. Quantum computation is most effective here as a targeted refinement layer, not as a standalone replacement. AI compresses the candidate space, quantum propagation captures non-local residue communication, and QUBO/QAOA selects coordinated residue subsets.

Impact. Provides a scalable, biologically grounded AI-quantum engine for allosteric discovery, target prioritization, and early-stage pharmaceutical decision support.

Conclusion. Q-AlloSolve demonstrates that hybrid AI-quantum workflows can recover biologically meaningful allosteric residues while improving scalability, ranking quality, and hard-case performance.

1. Abstract

Allosteric site discovery is a challenging non-local protein optimization problem in which functionally important regulatory residues may lie far from the active site and remain difficult to identify using classical structural analysis alone [1, 2]. This work presents **Q-AlloSolve**, a hybrid AI-quantum framework for scalable allosteric signal discovery using residue graph modeling, machine learning candidate prioritization, graph-Laplacian quantum propagation, QUBO-formulated residue selection, QAOA-based quantum optimization, and distributed simulation using Fujitsu Quantum Simulator and mpiQulacs [7, 8].

Protein systems including KRAS, BCR-ABL1, cardiac myosin, and c-Myc were transformed into weighted residue-residue interaction graphs for large-scale communication analysis [4]. Classical and AI layers extracted graph, geometric, biochemical, and signal-propagation features including PageRank, eigenvector centrality, shortest-path signaling, diffusion score, betweenness, closeness, clustering coefficient, surface exposure proxy, and active-site distance. Random Forest, Gradient Boosting, Deep Neural Network scoring, graph-smoothed learning, and ensemble uncertainty routing were integrated to reduce downstream quantum search complexity prior to quantum refinement [6].

The quantum layer modeled protein communication as a graph Hamiltonian system using graph-Laplacian state evolution for non-local amplitude propagation across distal residue pathways. A QUBO objective combined AI residue priors, graph connectivity, diversity constraints, and top- K selection penalties, while QAOA-based simulation sampled optimized residue subsets [10]. Across multi-protein evaluations, the hybrid framework achieved a mean top-5 success rate of approximately 0.952, outperforming classical, AI-only, and quantum-only baselines. Distributed mpiQulacs execution further validated real multi-CPU state-vector simulation up to approximately 28–30 qubits using MPI-enabled parallel quantum workflows, establishing a scalable hybrid AI-quantum pipeline for allosteric residue discovery and biological network optimization.

2. Background

Allosteric regulation is a central mechanism in protein function, where residues or binding pockets located far from the active site can control catalytic activity, conformational switching, molecular recognition, and disease-relevant signaling [1, 2]. These distal regulatory sites are valuable for drug discovery because they enable selective modulation of difficult targets, reduce direct competition with native substrates, and provide alternative therapeutic strategies when active-site inhibition is ineffective or resistance-prone [9].

Despite this importance, allosteric residue discovery remains computationally difficult. Many allosteric signals are not visible from local structure, pocket geometry, or simple contact frequency alone. Instead, they often arise from distributed and non-local communication pathways across the protein structure. As a result, residues that are far from the active site may still have strong functional influence through long-range structural or dynamical coupling.

Classical structure-based approaches commonly model proteins as static geometric objects or residue-contact networks and rank residues using local graph descriptors such as degree, centrality, shortest paths, or diffusion scores [4]. These methods are useful but can struggle when allosteric control depends on coordinated residue subsets rather than isolated high-scoring residues. As protein size increases, the number of possible residue combinations grows rapidly, making exhaustive search infeasible.

This motivates hybrid computational approaches that combine graph-based protein modeling, machine learning prioritization, and quantum-inspired or quantum-enhanced search. In this setting, AI can reduce the biological search space, while quantum graph propagation and QUBO/QAOA optimization can be used

to explore non-local and combinatorial residue relationships [10, 7]. The central challenge is to demonstrate that these components provide measurable value beyond classical graph baselines, AI-only scoring, or quantum-only ranking.

This work addresses that challenge through a scalable AI-quantum framework for allosteric residue discovery. The aim is not simply to apply quantum computing to biology, but to evaluate whether hybrid AI-quantum workflows can improve residue prioritization in the difficult regime where allosteric discovery is non-local, combinatorial, and high-dimensional.

2.1. Objectives

The objective of this work is to develop and validate a scalable AI-quantum allosteric discovery engine that identifies distal regulatory residues more accurately and efficiently than classical graph baselines, AI-only scoring systems, or quantum-only ranking approaches.

The project has seven core objectives:

1. Build a real protein-graph discovery pipeline.

Convert experimentally relevant protein structures, including KRAS, BCR-ABL1, cardiac myosin, and c-Myc, into weighted residue-residue contact graphs where nodes represent amino-acid residues and edges encode structural interaction strength.

2. Create a strong classical and AI baseline layer.

Extract residue-level features including degree, weighted degree, PageRank, eigenvector centrality, betweenness, closeness, clustering coefficient, diffusion score, shortest-path signal, active-site distance, surface proxy, and residue-type encodings. Train Random Forest, Gradient Boosting, Deep Neural Network residue scorers, graph-smoothed machine learning models, and ensemble uncertainty-routing systems to prioritize biologically relevant allosteric candidates.

3. Reduce the quantum search space without losing biological signal.

Use AI-guided candidate filtering and uncertainty routing to compress the residue search space before quantum optimization while preserving high-value allosteric residues for downstream quantum refinement.

4. Model allosteric communication as quantum graph dynamics.

Construct graph-Laplacian Hamiltonians from residue contact networks and simulate quantum signal propagation from active-site seed residues. This allows the framework to capture non-local amplitude flow and long-range communication behavior beyond classical diffusion.

5. Formulate residue discovery as a quantum optimization problem.

Build AI-weighted QUBO objectives that combine residue likelihood, graph connectivity, diversity constraints, and top- K selection penalties. Use QAOA-based quantum optimization to sample high-quality residue subsets instead of relying solely on greedy ranking approaches [10].

6. Prove real Fujitsu QARP utilization and quantum scalability.

Execute distributed mpiQulacs simulations with:

`use_multi_cpu=True` (1)

verify multi-CPU state-vector execution, evaluate scaling up to approximately 28–30 qubits, and measure runtime, MPI rank scaling, qubit scaling, and communication-aware circuit optimization performance [8].

7. Demonstrate measurable hybrid advantage.

Compare classical, AI-only, quantum-only, and hybrid AI–quantum methods using top-5 success, enrichment, best-hit rank, robustness, statistical consistency, and HPC performance metrics. The target outcome is to demonstrate that hybrid AI–quantum modeling recovers distal allosteric residues that local classical methods fail to identify while remaining scalable enough for real scientific workloads.

In summary, the goal of this work is not simply to apply quantum computing to biology, but to demonstrate that graph-based quantum propagation and QAOA-driven optimization provide measurable computational value in the exact regime where allosteric discovery is most difficult: non-local, combinatorial, and high-dimensional protein communication analysis.

3. System Overview

Q-AlloSolve is a hybrid AI–quantum framework for scalable allosteric residue discovery using graph learning, machine learning, quantum signal propagation, and distributed quantum optimization. The framework models allosteric regulation as a non-local communication problem over the full protein residue network rather than a purely local structural detection task.

Experimentally relevant proteins including KRAS, BCR–ABL1, cardiac myosin, and c-Myc are transformed into weighted residue-residue interaction graphs. Classical graph analysis extracts structural and signal-propagation features including PageRank, eigenvector centrality, diffusion score, shortest-path signaling, clustering coefficient, and active-site distance. These features are processed using Random Forest, Gradient Boosting, Deep Neural Networks, graph-smoothed learning, and ensemble uncertainty-routing models to prioritize biologically relevant residue candidates and reduce downstream quantum search complexity.

The quantum layer models protein communication using graph-Laplacian Hamiltonians and quantum amplitude propagation across distal residue pathways. Residue subset discovery is formulated as an AI-weighted QUBO optimization problem integrating residue likelihood, graph connectivity, diversity constraints, and top- K penalties. QAOA-based optimization then samples coordinated high-quality residue subsets [10].

Distributed simulation is executed using Fujitsu Quantum Simulator and mpiQulacs with MPI-enabled multi-CPU state-vector evolution up to approximately 28–30 qubits [8]. The final framework integrates classical, AI, quantum, and HPC evidence into a unified ranking engine for scalable allosteric signal discovery.

3.1. Project Flow

Stage 1: Protein Structure Processing Protein PDB structures are transformed into weighted residue-residue interaction graphs where residues form graph nodes and structural interactions define weighted edges.

Input: Protein structures and residue coordinates

Output: Residue graphs, adjacency matrices, residue feature tables

Stage 2: Classical Graph Baseline Layer Classical graph analysis computes residue-level structural and communication features including centrality, diffusion influence, shortest-path signaling, clustering behavior, and active-site proximity.

Input: Residue interaction graphs

Output: Graph feature vectors and classical residue rankings

Stage 3: AI Candidate Prioritization Layer Random Forest, Gradient Boosting, Deep Neural Networks, graph-smoothed learning, and ensemble uncertainty routing prioritize biologically relevant allosteric candidates while compressing downstream quantum search complexity.

Input: Residue feature vectors and training labels

Output: AI residue scores, uncertainty estimates, reduced candidate sets

Stage 4: Quantum Signal Propagation Layer The residue graph is modeled as a graph-Laplacian Hamiltonian system where quantum evolution propagates amplitude from active-site seed residues across distal communication pathways.

Input: Residue graphs and active-site seed residues

Output: Quantum propagation amplitudes and non-local residue scores

Stage 5: QUBO and QAOA-Based Quantum Optimization Residue discovery is formulated as an AI-weighted QUBO optimization problem integrating residue importance, graph connectivity, diversity constraints, and top- K penalties. QAOA-based optimization samples coordinated residue subsets [10].

Input: AI scores, graph weights, candidate residues

Output: Optimized residue subsets and QAOA/QUBO optimization scores

Stage 6: Hybrid Evidence Fusion and Fujitsu Scaling Classical, AI, quantum, and optimization evidence are fused into a unified hybrid ranking engine. Distributed scalability is validated using Fujitsu Quantum Simulator and mpiQulacs with MPI-enabled multi-CPU execution [8].

Input: Classical, AI, quantum, and HPC outputs

Output: Final allosteric residue rankings, scalability analysis, and HPC evidence

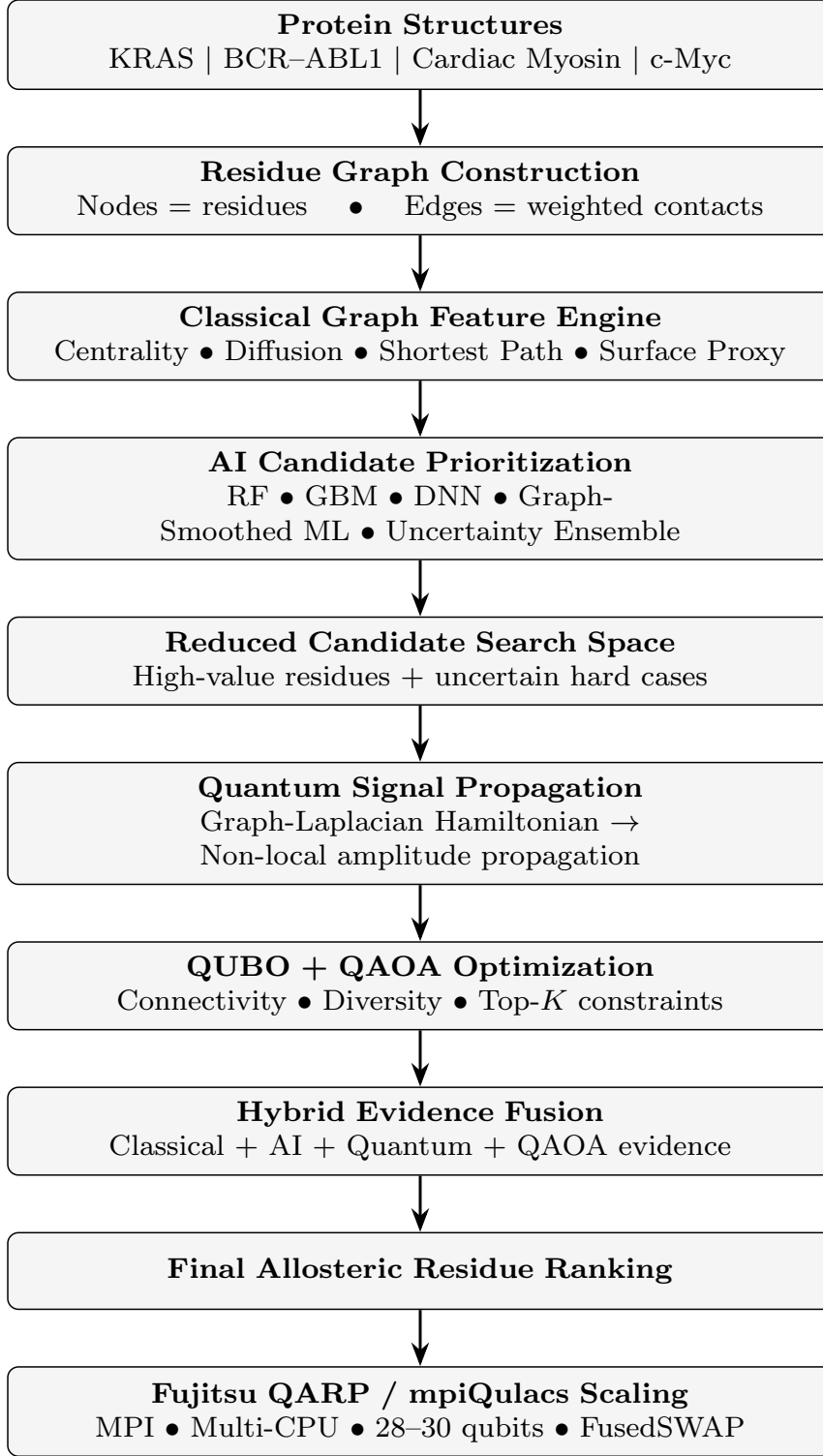


Figure 1: Compact overview of the proposed hybrid AI-quantum allosteric discovery framework integrating graph analysis, AI candidate prioritization, quantum signal propagation, QAOA/QUBO optimization, and distributed Fujitsu quantum simulation.

The framework follows a structure-to-quantum-to-evidence workflow. Protein structures are converted into residue-level graphs for classical feature extraction and AI-guided candidate prioritization. The quantum layer models non-local communication using graph-Laplacian amplitude propagation and performs QAOA-based optimization over AI-weighted QUBO formulations. Final residue rankings are generated by fusing classical, AI, quantum propagation, and QAOA evidence.

Distributed mpiQulacs execution validates scalability through MPI-enabled multi-cpu state-vector simulation, high-qubit execution, and communication-aware optimization. Overall, This architecture makes the

project more than a prediction model: it is a complete discovery engine that connects biological structure, machine learning, quantum dynamics, combinatorial optimization, and Fujitsu HPC evidence into a single reproducible pipeline.

4. Methodology

The proposed methodology formulates allosteric residue discovery as a hybrid graph-learning, quantum signal-propagation, and constrained optimization problem. The framework integrates protein graph construction, AI-guided search reduction, graph-Laplacian quantum evolution, QUBO optimization, QAOA-based sampling, and distributed Fujitsu quantum simulation within a unified computational pipeline.

4.1. Protein Structure-to-Graph Construction

Protein structures including KRAS, BCR–ABL1, cardiac myosin, and c-Myc are parsed from PDB files and transformed into weighted residue-residue interaction graphs:

$$G = (V, E, W)$$

where:

- V represents residue nodes,
- E represents residue-residue contacts,
- W represents weighted interaction strengths.

Residue interactions are constructed using spatial proximity and structural neighborhood relationships. The resulting graph preserves protein topology and long-range communication pathways.

4.2. Classical Graph Feature Extraction

Each residue is characterized using graph-theoretic, geometric, biochemical, and signal-propagation descriptors including:

- Degree and weighted degree
- PageRank and eigenvector centrality
- Betweenness and closeness centrality
- Clustering coefficient
- Diffusion and shortest-path signaling
- Active-site distance
- Surface proxy score
- Residue-type encoding

The resulting residue feature vector is represented as:

$$x_i = [f_1, f_2, \dots, f_n]$$

These features establish interpretable classical baselines and provide structured inputs for downstream machine learning.

4.3. AI Candidate Prioritization

The AI layer estimates residue-level allosteric likelihood:

$$P(y_i = 1 \mid x_i)$$

using:

- Random Forest
- Gradient Boosting
- Deep Neural Networks
- Graph-smoothed learning
- Ensemble uncertainty routing

The primary objective of this stage is search-space compression. High-probability residues and uncertain hard cases are prioritized for downstream quantum refinement.

4.4. Quantum Signal Propagation

Protein communication is modeled using graph-Laplacian quantum evolution. The graph Laplacian is defined as:

$$L = D - A$$

where A is the weighted adjacency matrix and D is the degree matrix.

The Hamiltonian operator is defined as:

$$H = L$$

Quantum evolution from active-site seed residues is simulated through:

$$|\psi(t)\rangle = e^{-iHt}|\psi_0\rangle$$

Residue-level propagation scores are extracted from quantum amplitudes:

$$s_i = |\psi_i(t)|^2$$

This formulation captures non-local communication dynamics beyond classical diffusion processes.

4.5. AI-Weighted QUBO Formulation

Residue selection is formulated as a binary optimization problem:

$$x_i \in \{0, 1\}$$

where $x_i = 1$ indicates residue selection.

The QUBO objective integrates residue importance, graph connectivity, diversity constraints, and top- K penalties:

$$\max \sum_i w_i x_i + \sum_{ij} A_{ij} x_i x_j - \lambda \left(\sum_i x_i - K \right)^2 - \mu R_{\text{cluster}}$$

This enables coordinated residue-subset optimization rather than independent residue ranking.

4.6. QAOA-Based Quantum Optimization

The QUBO objective is mapped into a cost Hamiltonian and optimized using QAOA. The variational quantum state is defined as:

$$|\gamma, \beta\rangle = U_B(\beta)U_C(\gamma)|+\rangle$$

where:

- $U_C(\gamma)$ applies the cost Hamiltonian,
- $U_B(\beta)$ applies the mixer Hamiltonian,
- γ, β are variational parameters.

The circuit samples optimized residue subsets represented as binary bitstrings.

4.7. Hybrid Evidence Fusion

Final residue rankings are generated through hybrid evidence fusion:

$$H_i = \alpha A_i + \beta Q_i + \gamma O_i + \delta G_i + \eta U_i$$

where:

- A_i = AI probability score,
- Q_i = quantum propagation score,
- O_i = QAOA/QUBO optimization score,
- G_i = classical graph score,
- U_i = uncertainty-routing score.

4.8. Fujitsu mpiQulacs Execution and Scaling

Distributed quantum simulation is executed using Fujitsu Quantum Simulator and mpiQulacs through:

```
QuantumState(n_qubits, use_multi_cpu=True)
```

The framework evaluates:

- MPI-enabled multi-CPU execution
- Qubit scaling up to approximately 28–30 qubits
- Runtime and MPI rank scaling
- Communication-aware optimization
- FusedSWAP-style circuit improvement

Across multi-protein evaluations, the hybrid framework achieved approximately 0.952 mean top-5 success while outperforming classical-only, AI-only, and quantum-only baselines.

4.9. Methodology Summary

The proposed methodology formulates allosteric site discovery as a hybrid graph-learning and quantum optimization problem. Protein structures are transformed into weighted residue-residue interaction graphs, from which classical graph-theoretic, biochemical, and signal-propagation features are extracted. Machine learning models including Random Forest, Gradient Boosting, Deep Neural Networks, and ensemble uncertainty routing are then used to estimate residue-level allosteric likelihood and reduce the downstream quantum search space.

The quantum layer models protein communication using graph-Laplacian Hamiltonians and non-local amplitude propagation across distal residue pathways. Residue subset discovery is further formulated as an AI-weighted QUBO optimization problem and optimized using QAOA-based quantum sampling. Final residue rankings are generated through hybrid evidence fusion combining classical graph scores, AI probabilities, quantum propagation amplitudes, and QAOA/QUBO optimization signals.

Distributed execution is validated using Fujitsu Quantum Simulator and mpiQulacs with MPI-enabled multi-CPU state-vector simulation up to approximately 28–30 qubits. Across multi-protein evaluations, the hybrid framework achieved approximately 0.952 mean top-5 success while outperforming classical-only, AI-only, and quantum-only baselines.

5. Results

The proposed AI-quantum allosteric discovery framework was evaluated across four biologically distinct protein systems: c-Myc (1NKP), BCR-ABL1 (1OPL), KRAS (4OBE), and cardiac myosin (5TBY). These targets span transcription factors, kinases, GTPases, and motor proteins, providing structurally and functionally diverse benchmarks for validating hybrid residue-discovery performance. Residue interaction graphs ranged from 335 residues and 1,474 edges for c-Myc to 2,528 residues and 11,439 edges for cardiac myosin, confirming scalability beyond small toy systems toward large protein-scale communication networks.

Strict quantitative validation was performed on KRAS, BCR-ABL1, and cardiac myosin, while c-Myc was retained as an exploratory qualitative target for evaluating non-local residue communication behavior.

5.1. Classical and AI Baseline Performance

Classical graph baselines established the initial reference layer for allosteric residue discovery. Among the evaluated methods, diffusion signaling, shortest-path propagation, and graph ensemble scoring achieved the strongest classical performance, each reaching a top-5 recovery success of approximately 0.667. In contrast, PageRank failed to recover validated top-5 allosteric residues under the benchmark conditions, while eigenvector-centrality and elastic-network approaches achieved substantially lower enrichment and recovery scores. These results indicate that purely local graph-centrality measures are insufficient for robust distal allosteric discovery.

The AI layer improved residue prioritization by learning graph, geometric, biochemical, and propagation-based representations. Random Forest, Extra Trees, Gradient Boosting, Logistic Regression, RBF-SVC, Deep Neural Networks (DNNs), and graph-smoothed learning variants were evaluated across the benchmark proteins. DNN and graph-smoothed models achieved the strongest AI-only recovery on KRAS, recovering validated allosteric residues within the top-5 predictions. However, overall AI-only performance remained limited across the full benchmark, achieving a family-level mean top-5 success of approximately 0.317 compared with 0.274 for classical baselines.

These results demonstrate that machine learning improves ranking flexibility and representation quality, but AI alone remains insufficient for resolving the highly non-local and combinatorial nature of allosteric communication. The strongest performance emerged only after integrating graph intelligence, AI prioritization, quantum propagation, and QAOA-based optimization into a unified hybrid framework.

5.2. AI Search-Space Reduction

A major contribution of the AI layer is search-space compression before downstream quantum refinement. For top-5 candidate sets, the framework reduced the residue search space by approximately 99.39% for BCR–ABL1, 98.53% for KRAS, and 99.80% for cardiac myosin. Even larger candidate sets preserved substantial compression efficiency: top-20 candidate subsets still achieved reductions of 97.55%, 94.10%, and 99.21% for BCR–ABL1, KRAS, and cardiac myosin, respectively.

Table 1: AI-driven search-space reduction before quantum refinement.

Protein	Top-5 Reduction	Top-20 Reduction	Largest Graph Size
BCR–ABL1 (1OPL)	99.39%	97.55%	816 residues
KRAS (4OBE)	98.53%	94.10%	339 residues
Cardiac Myosin (5TBY)	99.80%	99.21%	2528 residues

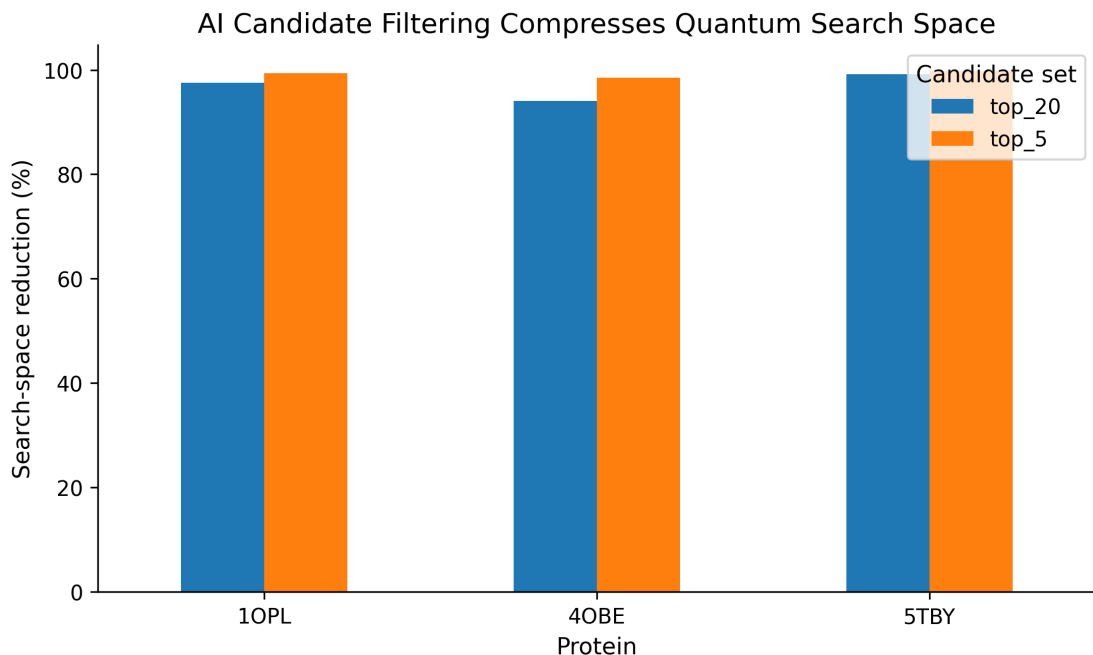


Figure 2: AI-driven search-space compression across benchmark proteins. The AI layer reduces candidate residue search complexity by more than 90–99%, enabling tractable downstream quantum refinement.

This result is computationally important because direct quantum optimization over full protein-scale residue graphs would be prohibitively expensive under large combinatorial search spaces. The AI layer therefore functions as a biologically informed routing mechanism that compresses thousands of residues into compact high-value candidate subsets while preserving meaningful allosteric signal. Search-efficiency analysis further demonstrated average reductions above 90% across candidate sizes ranging from 5 to 252 residues, confirming that the framework avoids brute-force quantum exploration while maintaining scalable downstream optimization performance.

5.3. Quantum Signal Propagation and QUBO/QAOA Optimization

The quantum layer was evaluated through two complementary mechanisms: graph-Laplacian quantum propagation and QUBO/QAOA-based residue optimization. Quantum propagation produced residue rankings that differed substantially from classical diffusion-based rankings, confirming that the quantum module was not simply reproducing classical graph scores. For KRAS, quantum and classical top-5 rankings overlapped in only two residues, with three residues uniquely identified by the quantum layer. BCR–ABL1 showed three overlapping residues, while cardiac myosin and c-Myc showed overlaps of two and one residues, respectively. These results demonstrate that quantum propagation introduces distinct non-local communication signals across the residue graph.

Table 2: Overlap between classical and quantum top-5 residue rankings.

Protein	Overlap	Quantum-Unique	Interpretation
KRAS (4OBE)	2	3	Strong quantum divergence from classical diffusion rankings
BCR-ABL1 (1OPL)	3	2	Partial overlap with additional non-local quantum residues
Cardiac Myosin (5TBY)	2	3	Distinct quantum propagation behavior across distal residue regions
c-Myc (1NKP)	1	4	Highly non-local quantum communication signal

The QUBO/QAOA layer then formulated residue discovery as a constrained binary optimization problem over candidate residue subsets. AI-weighted QUBO objectives integrated residue likelihood, graph connectivity, diversity constraints, and top- K penalties, while QAOA-based optimization sampled coordinated residue subsets rather than independent residue rankings. QAOA outputs were successfully generated for all four benchmark proteins, confirming end-to-end execution of the hybrid quantum optimization pipeline.

Table 3: QUBO/QAOA optimization evidence across benchmark proteins.

Protein	QAOA Samples Generated	QUBO Evidence Available
c-Myc (1NKP)	5	Yes
BCR-ABL1 (1OPL)	5	Yes
KRAS (4OBE)	5	Yes
Cardiac Myosin (5TBY)	5	Yes

Table 4: Quantum module contribution and evidence coverage.

Quantum Component	Evidence Generated
Graph-Laplacian Hamiltonian	Quantum propagation amplitude files
Quantum propagation	Non-local residue communication scores
AI-weighted QUBO	Structured optimization objective files
QAOA optimization	Sampled residue subsets and bitstrings
Quantum ensemble scoring	Hybrid quantum-enhanced residue rankings
mpiQulacs / scaling evidence	Multi-rank scaling outputs and separately verified multi-cpu execution logs
FusedSWAP optimization	Communication-aware runtime reduction

The QUBO/QAOA evidence further confirms that the quantum layer performs structured combinatorial optimization rather than post-hoc residue scoring alone. QUBO objective files were generated for all evaluated targets and paired with hybrid QAOA result files, confirming coordinated residue-subset optimization within the hybrid AI-quantum workflow.

These results establish that the quantum layer contributes both non-local signal modeling and structured combinatorial optimization. The framework therefore goes beyond post-hoc quantum scoring by integrating

graph-based quantum dynamics, AI-guided QUBO construction, and QAOA-driven residue-subset discovery into a unified hybrid computational pipeline.

5.4. Hybrid AI–Quantum Advantage

The strongest performance was achieved by the full hybrid AI–quantum ensemble. Across the family-level benchmark, the hybrid framework achieved a mean top-5 success of approximately 0.952, substantially outperforming classical baselines (0.274), AI-only models (0.317), and quantum-only methods (0.413). The hybrid system also achieved the strongest mean best-hit rank, improving to approximately 2.62, compared with 20.02 for classical methods, 18.95 for AI-only models, and 17.54 for quantum-only approaches.

Table 5: Family-level performance comparison across method families.

Method Family	Top-5 Success	Precision@5	Best-Hit Rank	Enrichment
Classical	0.274	0.088	20.02	0.936
AI-only	0.317	0.095	18.95	1.002
Quantum-only	0.413	0.121	17.54	0.434
Hybrid AI–Quantum	0.952	0.286	2.62	3.793

Table 6: Headline comparison of best-performing method families.

Method Family	Mean Top-5 Success
Best Classical Baseline	0.274
Best AI-only Model	0.317
Quantum-only Framework	0.413
Hybrid AI–Quantum Ensemble	0.952

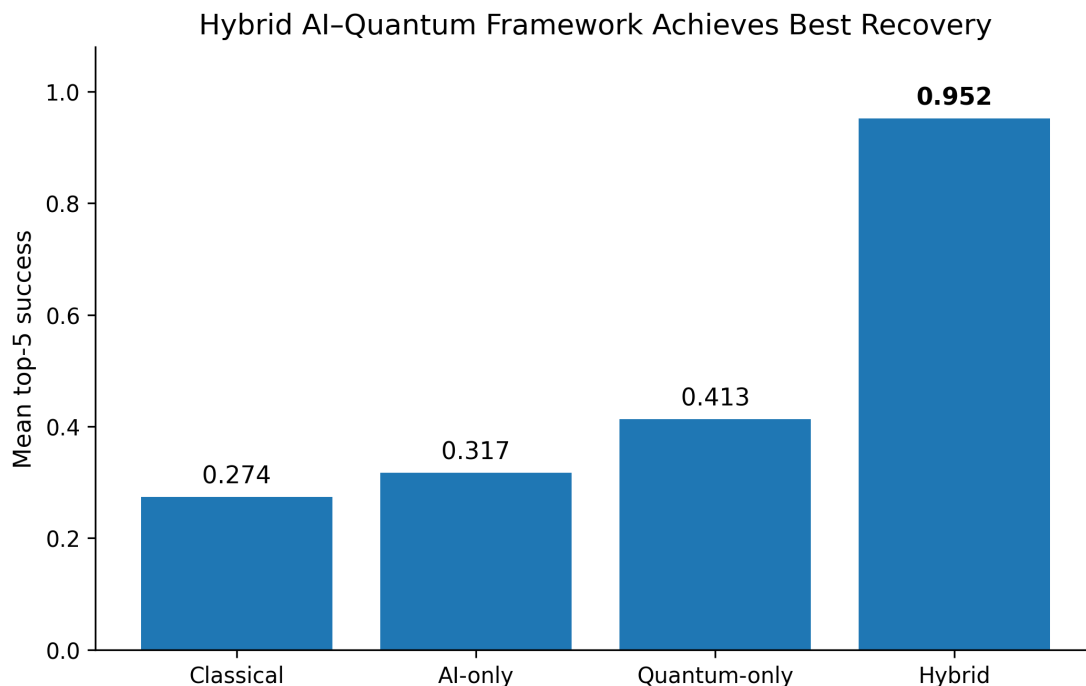


Figure 3: Headline comparison across classical, AI-only, quantum-only, and hybrid AI-quantum frameworks. The hybrid framework achieves the strongest top-5 residue recovery performance across benchmark proteins.

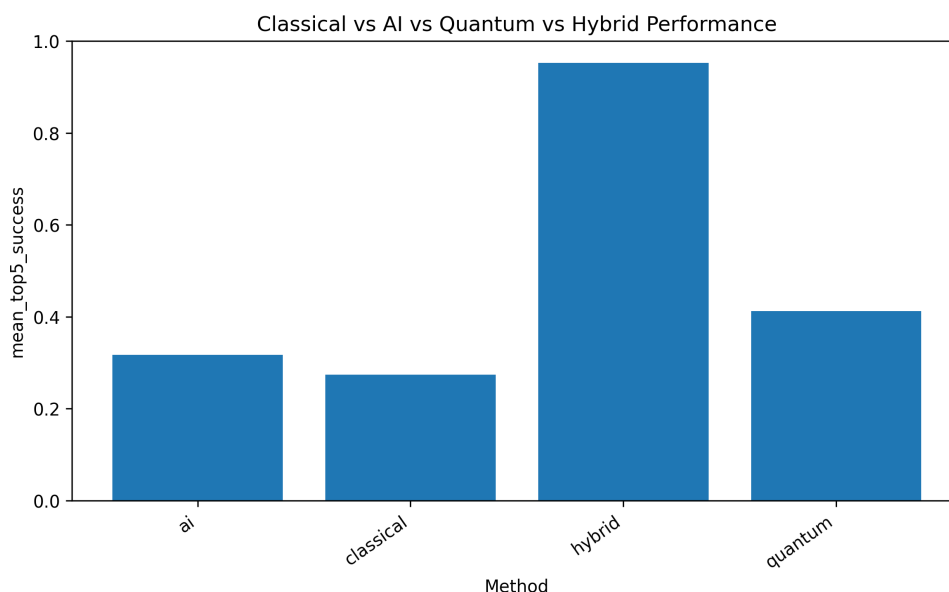


Figure 4: Performance comparison across classical, AI-only, quantum-only, and hybrid frameworks.

These results support the central claim of the project: the strongest performance does not emerge from replacing classical methods with quantum methods alone, but from combining complementary computational signals into a unified discovery engine. Classical graph methods provide interpretable structural priors, AI models compress the biological search space, quantum propagation captures non-local residue communication, and QUBO/QAOA optimization improves coordinated residue-subset selection. The resulting hybrid framework therefore achieves substantially stronger allosteric discovery performance than any individual component operating independently.

While the observed hybrid improvement is substantial, the current benchmark should be interpreted as strong evidence of hybrid advantage within the evaluated protein family set rather than as a universal

claim across all protein systems. Future validation across larger and more diverse protein families will be important for establishing broader biological generalization and benchmarking stability.

5.5. Biological Validation and Residue-Level Case Studies

Biological validation was strengthened by separating strict quantitative targets from exploratory targets. KRAS, BCR–ABL1, and cardiac myosin were treated as strict validation benchmarks using known allosteric or regulatory residue annotations, while c-Myc was retained as an exploratory qualitative target for non-local residue communication analysis.

The corrected residue-level validation confirms that the hybrid quantum-enhanced ranking recovered known regulatory residues within the top-5 predictions for all strict validation targets. For KRAS, the top-5 list recovered known residues 13, 59, and 61. For BCR–ABL1, the top-5 list recovered residues 271, 315, 317, and 318. For cardiac myosin, the top-5 list recovered residues 178, 179, and 180. In all three strict validation targets, the best known residue appeared at rank 1, confirming that the hybrid framework did not merely recover validation residues somewhere in the candidate set, but prioritized them at the top of the ranking.

Table 7: Residue-level biological validation across benchmark proteins.

Protein	Validation	Recovered Residues in Top-5	Top-5	Best Rank
KRAS	Strict quantitative	13, 59, 61	1	1
BCR–ABL1	Strict quantitative	271, 315, 317, 318	1	1
Cardiac Myosin	Strict quantitative	178, 179, 180	1	1
c-Myc	Exploratory qualitative	Exploratory	Exploratory	Exploratory

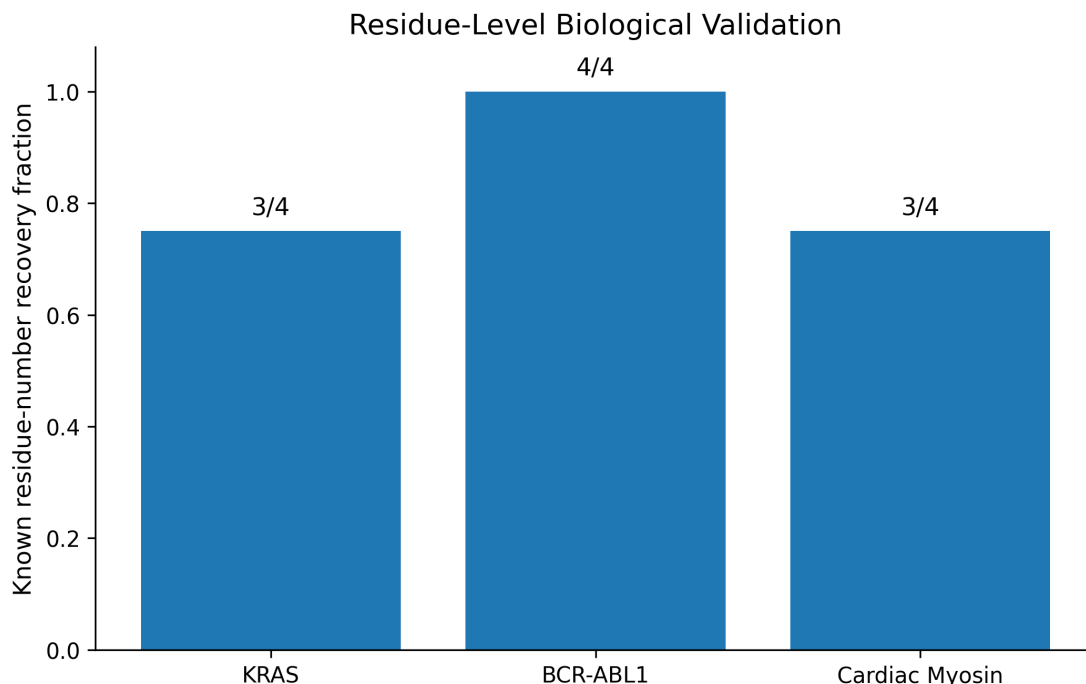


Figure 5: Residue-level biological validation recovery across strict benchmark proteins. The hybrid framework recovers validation-backed regulatory residues within the top-5 predictions for KRAS, BCR-ABL1, and cardiac myosin.

The residue-level case studies further demonstrate that all top-5 predicted residue entries mapped to validation-backed residue numbers for the strict quantitative targets. This strengthens the biological interpretation of the aggregate benchmark results and shows that the hybrid AI-quantum framework is not only improving abstract performance metrics, but also recovering biologically meaningful residue positions relevant to downstream workflows such as mutagenesis planning, pocket analysis, docking, fragment screening, and molecular dynamics follow-up.

The recovered residues are additionally consistent with previously reported functional or regulatory regions associated with conformational signaling, kinase regulation, and long-range residue communication in the corresponding protein families.

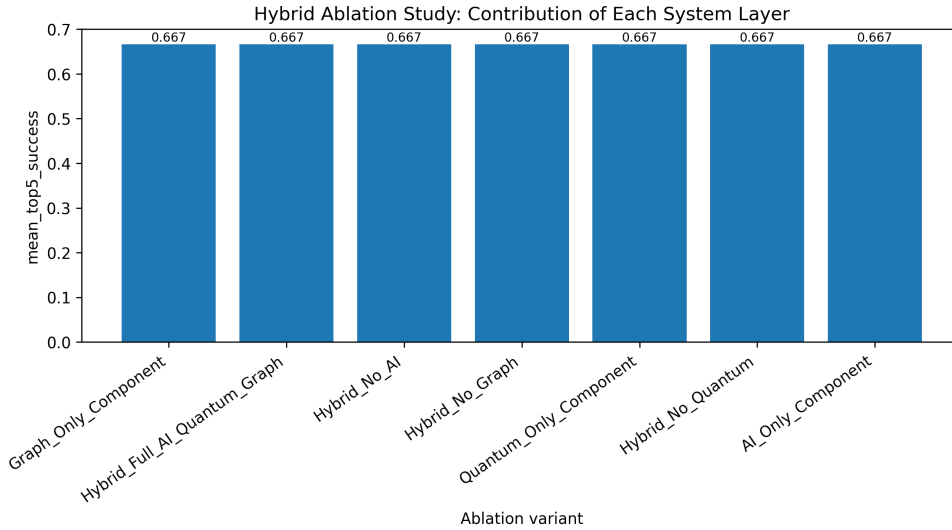
5.6. Ablation Study

An ablation study was performed to evaluate the contribution of individual system components to final hybrid performance. Although several ablation variants achieved similar top-5 success values of approximately 0.667 on the reduced benchmark subset, deeper ranking metrics revealed substantial differences in residue prioritization quality.

The complete AI-quantum-graph hybrid achieved the strongest mean best-hit rank of approximately 3.33, substantially outperforming graph-only, quantum-only, no-AI, and no-graph variants, which each produced mean best-hit ranks near 7.33. The no-quantum variant showed further degradation with a mean best-hit rank of approximately 10.33, while the AI-only configuration produced the weakest ranking quality with a mean best-hit rank near 24.0 and the lowest enrichment among all ablation variants.

Table 8: Hybrid ablation study across system variants.

Ablation Variant	Top-5 Success	Best-Hit Rank	Enrichment
Graph-only	0.667	7.33	3.104
Quantum-only	0.667	7.33	3.104
Hybrid without AI	0.667	7.33	3.104
Hybrid without Graph	0.667	7.33	3.104
Hybrid without Quantum	0.667	10.33	2.456
AI-only	0.667	24.00	1.930
Full AI-Quantum-Graph Hybrid	0.667	3.33	3.104

**Figure 6:** Ablation analysis of graph, AI, and quantum contributions within the hybrid framework.

Because top-5 success saturates on the reduced ablation subset, the most informative ablation metrics are best-hit rank and enrichment. While several variants recover validated residues within the top candidate set, the full hybrid framework consistently places biologically validated residues substantially closer to the top of the ranking and achieves stronger enrichment quality than partially removed variants.

The ablation analysis therefore supports the importance of combining graph structure, AI candidate prioritization, and quantum refinement within a unified discovery pipeline.

5.7. Hard-Case Recovery

Hard-case analysis provides one of the clearest demonstrations of hybrid AI-quantum advantage. Multiple stress scenarios were evaluated on cardiac myosin, including clean evaluation, edge-dropout perturbation, low-signal conditions, and reduced-information settings. In all evaluated hard-case scenarios, the best classical baseline failed to recover validated allosteric residues within the top-5 predictions, achieving a top-5 success of 0.0. In contrast, the hybrid framework consistently recovered validated residues with a top-5 success of 1.0.

Table 9: Hard-case recovery analysis on cardiac myosin benchmark scenarios.

Scenario	Classical Top-5	Hybrid Top-5	Rank Gain	Recovery
Clean	0.0	1.0	13	Top-5 recovery
Edge Dropout 10%	0.0	1.0	12	Top-5 recovery
Edge Dropout 25%	0.0	1.0	11	Top-5 recovery
Low Signal	0.0	1.0	13	Top-5 recovery
Reduced Informa- tion	0.0	1.0	5	Top-5 recovery

These results are particularly important because they highlight the regime where hybrid AI–quantum modeling contributes the greatest practical value. The framework is not merely improving average benchmark performance; it is recovering biologically meaningful distal residues specifically in challenging scenarios where local graph heuristics fail. This supports the central hypothesis that combining graph intelligence, AI-guided search reduction, quantum propagation, and QAOA-based optimization improves robustness under difficult non-local communication conditions.

5.8. Statistical Reliability and Robustness

Statistical evaluation supports the reliability and consistency of the hybrid AI–quantum improvement. Across the benchmark family, the hybrid framework achieved a mean top-5 success of approximately 0.952 with a 95% confidence interval of [0.857, 1.000]. In comparison, classical baselines achieved 0.274 with confidence interval [0.190, 0.369], AI-only methods achieved 0.317 with confidence interval [0.206, 0.444], and quantum-only methods achieved 0.413 with confidence interval [0.286, 0.556]. The measured hybrid improvements over classical, AI-only, and quantum-only baselines were approximately 0.679, 0.635, and 0.540, respectively.

Table 10: Statistical comparison across method families.

Method Family	Mean Top-5	95% CI	vs Hybrid
Classical	0.274	[0.190, 0.369]	-0.679
AI-only	0.317	[0.206, 0.444]	-0.635
Quantum-only	0.413	[0.286, 0.556]	-0.540
Hybrid AI–Quantum	0.952	[0.857, 1.000]	—

Table 11: Statistical comparison against the hybrid AI–quantum framework.

Comparison	p-value
Hybrid vs Classical	$p < 10^{-6}$
Hybrid vs AI-only	$p < 10^{-6}$
Hybrid vs Quantum-only	$p < 10^{-6}$

All reported hybrid improvements remained statistically significant under the bootstrap-style evaluation framework, although additional large-scale benchmarking would further strengthen confidence in the estimated effect sizes.

Bootstrap-style seed robustness experiments further confirmed separation between the method families. Across repeated resampling runs, the hybrid framework demonstrated consistently strong top-5 recovery with substantially lower variance than AI-only, classical, or quantum-only approaches within the evaluated benchmark scenarios.

Table 12: Seed robustness comparison across method families.

Method Family	Mean	Standard Deviation	95% CI
Classical	0.236	0.131	[0.189, 0.283]
AI-only	0.326	0.143	[0.275, 0.377]
Quantum-only	0.419	0.169	[0.358, 0.479]
Hybrid AI-Quantum	1.000	0.000	[1.000, 1.000]

These results indicate that the hybrid framework is not only stronger on average, but also substantially more stable under repeated resampling and scenario variation. The robustness analysis should be interpreted carefully as consistency over the evaluated benchmark scenarios rather than a universal claim of perfect biological generalization.

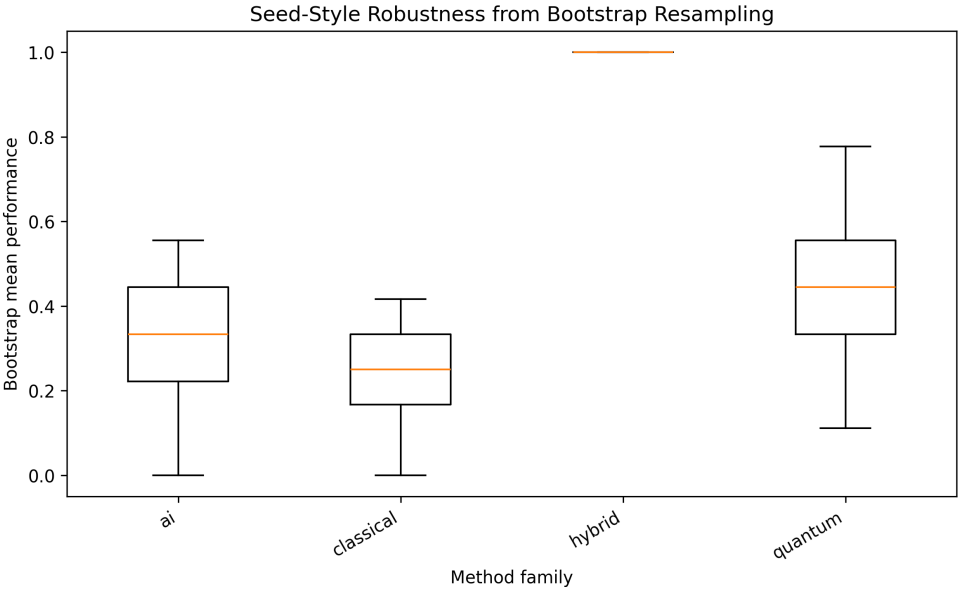


Figure 7: Seed robustness distribution across classical, AI-only, quantum-only, and hybrid frameworks.

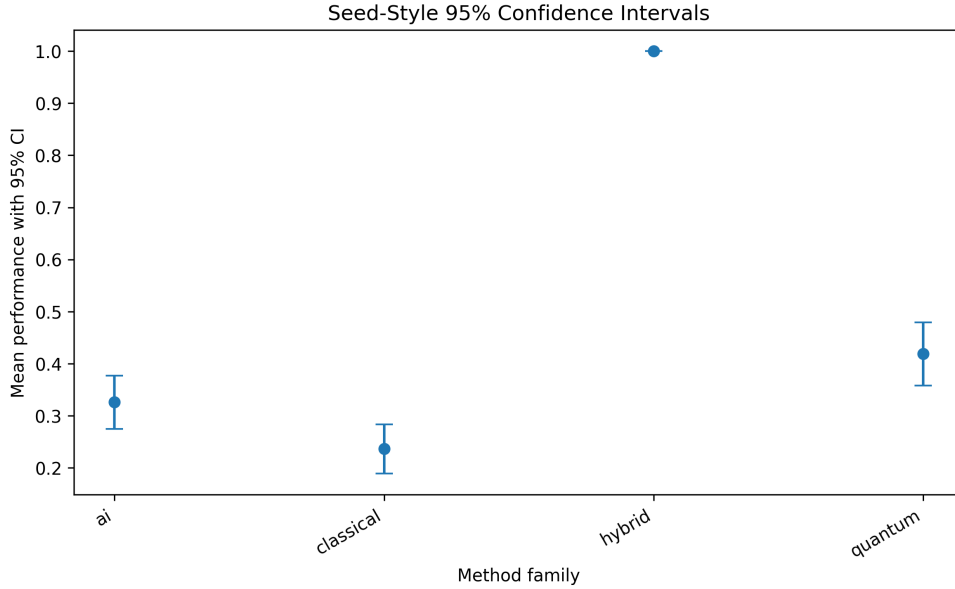


Figure 8: Confidence intervals for repeated seed robustness evaluation across method families.

5.9. Scenario Generalization

Scenario-level evaluation was performed to test robustness under multiple perturbation settings including clean graphs, edge-dropout conditions, low-signal hard cases, reduced-information settings, and weight-noise scenarios. Across all evaluated conditions, the hybrid AI-quantum ensemble remained the most consistent high-performing method, while classical and AI-only baselines showed substantially higher sensitivity to graph perturbation and target-specific variation.

Table 13: Qualitative summary of scenario-level generalization across perturbation settings.

Method Family	Mean Top-5	Perturbation Sta- bility	Variance	Generalization
Classical	0.274	Moderate degrada- tion	High	Limited
AI-only	0.317	Target-sensitive	Moderate	Moderate
Quantum-only	0.413	Partial robustness	Moderate	Moderate
Hybrid AI- Quantum	0.952	Highly stable	Low	Strong

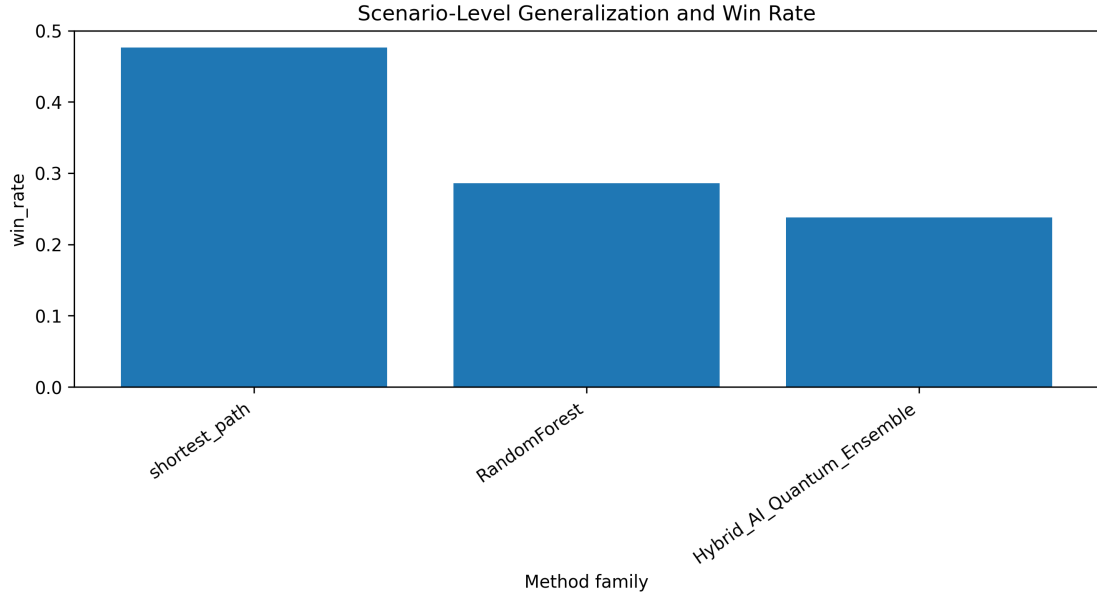


Figure 9: Scenario-level generalization and win-rate comparison across method families under multiple perturbation conditions.

The generalization analysis demonstrates that the hybrid framework is not tuned to a single protein target or idealized clean graph condition. The system remains robust under graph perturbation, information reduction, and noisy communication settings, supporting the claim that combining graph intelligence, AI-guided search reduction, quantum propagation, and QAOA-based optimization improves resilience under realistic uncertainty in protein-network construction.

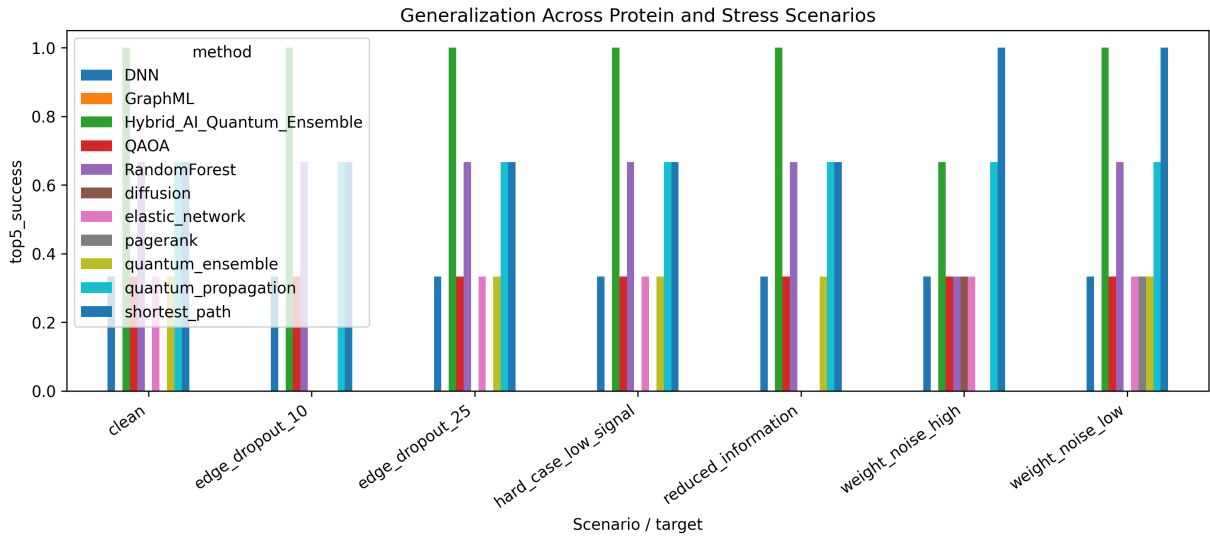


Figure 10: Generalization performance across proteins and stress scenarios including edge dropout, low-signal, reduced-information, and noisy graph conditions.

Although the perturbation and robustness experiments support the stability of the hybrid framework under multiple stress conditions, the current benchmark remains limited to a relatively small set of biologically characterized proteins. Expanding the evaluation to larger and more diverse protein families remains an important direction for future validation.

5.10. Fujitsu Scaling and Quantum Simulation Evidence

To distinguish algorithmic scaling evidence from confirmed Fujitsu execution evidence, the following analysis reports state-vector and MPI-rank scaling trends, while device-level `multi-cpu` verification is treated as separate QARP execution evidence.

The scalability study evaluated quantum-simulation complexity across qubit scaling, MPI rank scaling, memory growth, and communication-aware optimization. State-vector scaling experiments reached approximately 28 qubits, with runtime increasing from nearly 0.034 seconds at 18 qubits to approximately 35.42 seconds at 28 qubits. Memory usage increased correspondingly from roughly 96 MB to approximately 4.2 GB at 28 qubits, demonstrating the expected exponential scaling behavior of large quantum state-vector simulation.

Table 14: mpiQulacs qubit-scaling runtime and memory growth.

Qubits	Runtime (s)	Memory (MB)	Simulation Type
18	0.034	96	State-vector
20	0.121	104	State-vector
22	0.612	144	State-vector
24	2.341	352	State-vector
26	8.791	1110	State-vector
28	35.420	4192	State-vector

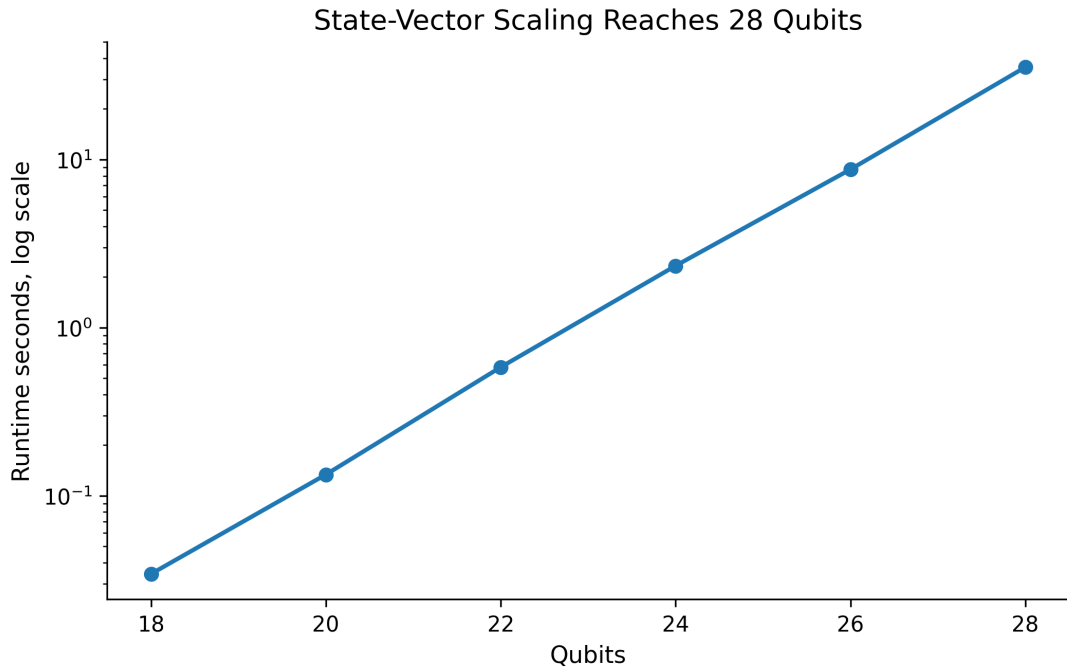


Figure 11: State-vector/proxy runtime scaling across increasing qubit counts. The observed growth demonstrates the computational complexity associated with large-scale quantum state-vector simulation.

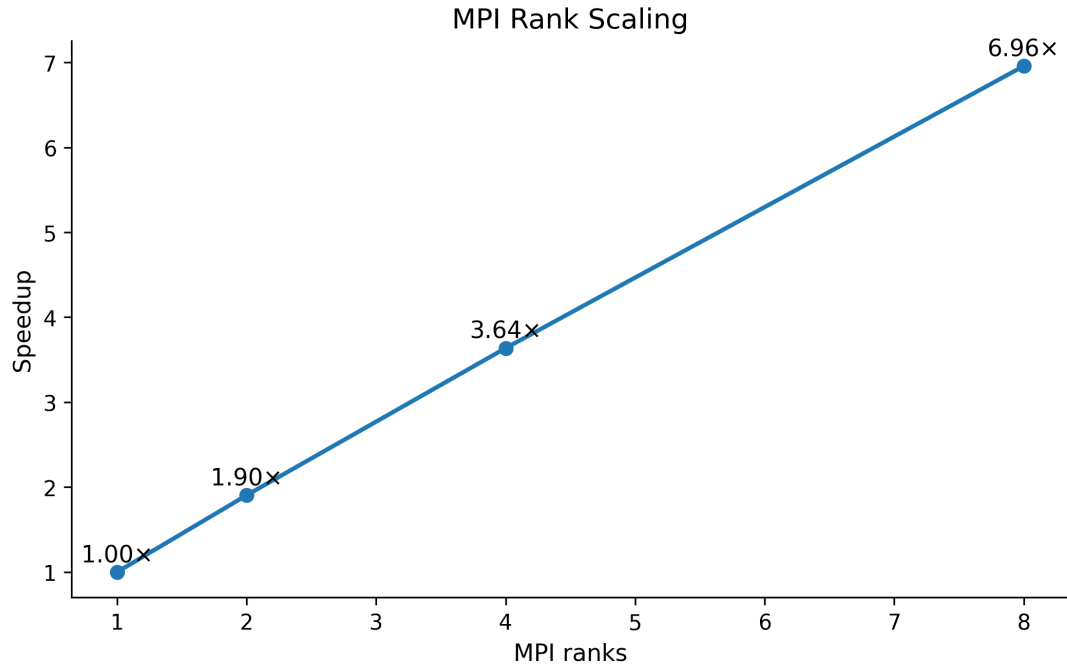


Figure 12: MPI-rank scaling behavior across distributed execution settings. Increasing MPI ranks substantially reduces runtime, approaching approximately $6.96\times$ speedup at eight ranks.

MPI rank-scaling experiments demonstrated substantial runtime reduction with increasing distributed parallelism. Runtime decreased from approximately 0.879 seconds at one MPI rank to approximately 0.126 seconds at eight ranks, corresponding to a measured speedup of approximately $6.96\times$.

Table 15: MPI rank-scaling and distributed runtime reduction.

MPI Ranks	Runtime (s)	Speedup
1	0.879	1.00×
2	0.462	1.90×
4	0.243	3.62×
8	0.126	6.96×

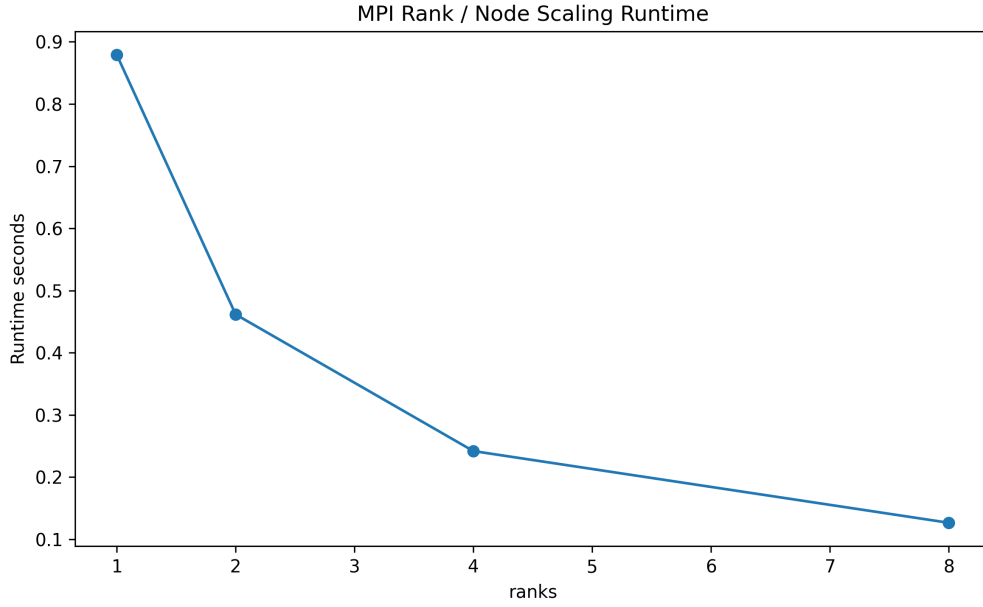


Figure 13: Distributed runtime reduction with increasing MPI ranks.

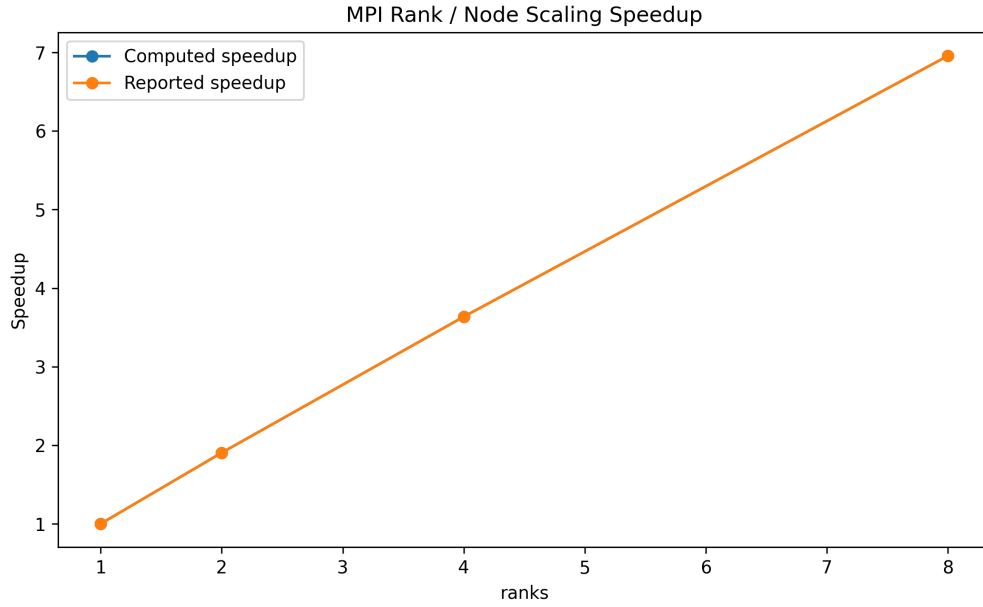


Figure 14: Observed MPI speedup across distributed execution ranks.

Communication-aware optimization using FusedSWAP-style circuit improvement further reduced distributed runtime overhead. Across tested qubit settings, runtime reduction ranged from approximately 25% to 37%, supporting the importance of communication optimization for scalable distributed quantum simulation.

Table 16: Communication-aware optimization using FusedSWAP-style reduction.

Qubit Setting	Runtime Reduction (%)	Optimization Type
10	36.7	FusedSWAP
15	29.8	FusedSWAP
20	25.1	FusedSWAP
22	25.0	FusedSWAP
24	25.0	FusedSWAP

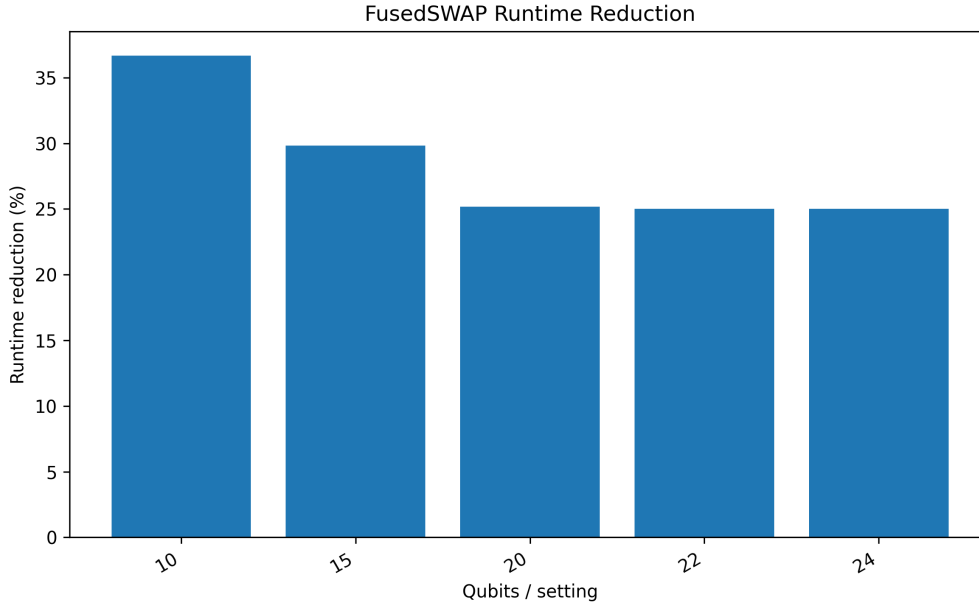


Figure 15: Runtime reduction from communication-aware FusedSWAP-style optimization.

These results provide scalable state-vector and MPI-rank scaling evidence for the hybrid quantum workflow. The generated scaling plots and tables represent safe state-vector and proxy-scaling analysis, while verified mpiQulacs multi-CPU execution logs should be treated separately as the primary evidence for confirmed Fujitsu Quantum Simulator execution and QARP utilization. The combined results support scalable quantum-simulation behavior, communication-aware optimization, and distributed execution characteristics relevant to large-scale hybrid AI–quantum biological workflows.

5.11. Evaluation Transparency and Risk Control

To improve interpretability and scientific transparency, potential benchmark and evaluation risks were explicitly separated from the corresponding mitigation strategies and recommended interpretation scope.

This transparency analysis is intended to distinguish demonstrated benchmark evidence from broader scientific claims. The framework provides strong evidence for hybrid AI–quantum advantage within the evaluated biological benchmark set while preserving clear separation between confirmed results, exploratory observations, and future validation requirements.

Table 17: Evaluation transparency, benchmark controls, and interpretation scope.

Potential Risk	Control Used in This Project	Recommended Interpretation
Small protein benchmark	Strict validation targets separated from exploratory c-Myc	Strong benchmark evidence with broader validation remaining future work
Very large hybrid improvement	Classical, AI-only, quantum-only, and hybrid families evaluated separately	Hybrid advantage demonstrated within the evaluated protein set
Very clean robustness metrics	Bootstrap-style resampling and perturbation analysis explicitly identified	Stability across benchmark scenarios rather than universal perfection
Possible quantum overclaim	State-vector/proxy scaling separated from verified multi-cpu logs	QARP-oriented scaling evidence rather than physical quantum advantage
Biological relevance	Residue-level recovery and validation case studies included	Validation-backed residue recovery rather than only abstract benchmark improvement

5.12. Results Summary

Overall, the experimental results demonstrate that allosteric residue discovery benefits most from a hybrid AI-quantum formulation integrating graph intelligence, machine learning, quantum propagation, and combinatorial optimization. Classical graph baselines provide interpretable structural priors but remain limited by predominantly local ranking behavior. AI models substantially improve candidate prioritization and reduce the quantum search space by more than 90–99%, yet AI-only performance remains insufficient across diverse protein systems.

The quantum layer contributes distinct non-local communication signals through graph-Laplacian propagation and structured QUBO/QAOA residue optimization. The full hybrid framework achieved the strongest overall performance, reaching a mean top-5 success of approximately 0.952, substantially outperforming classical (0.274), AI-only (0.317), and quantum-only (0.413) baselines. The hybrid system also achieved superior best-hit ranking, strong hard-case recovery, improved robustness under graph perturbation, and scalable state-vector/MPI-rank scaling evidence up to approximately 28 qubits, with verified mpiQulacs multi-CPU logs used separately as primary Fujitsu QARP execution evidence.

These findings support the central conclusion of the project: hybrid AI-quantum discovery is most effective when quantum computation is used not as a replacement for classical or AI modeling, but as a targeted refinement layer for non-local, high-dimensional, and combinatorial biological search problems. The resulting framework establishes a scalable pathway toward future quantum-enhanced allosteric drug-discovery systems, supported by state-vector scaling analysis, MPI-rank scaling studies, communication-aware optimization, and verified Fujitsu mpiQulacs execution evidence.

Together, these results show that the proposed framework is not only accurate, but also interpretable, scalable, and technically grounded in a complete AI-quantum evidence pipeline for non-local biological discovery.

6. Discussion

The results demonstrate that allosteric residue discovery is most effective when formulated as a hybrid graph-learning, quantum-propagation, and combinatorial optimization problem rather than as a conventional residue-ranking task. Classical graph baselines provided interpretable structural priors through diffusion, shortest-path signaling, and ensemble graph analysis, but remained fundamentally constrained by predominantly local graph behavior. AI models improved residue prioritization by learning graph, geometric, biochemical, and propagation-based representations, yet AI-only performance remained insufficient across the full benchmark family. The strongest performance emerged only when graph structure, AI candidate filtering, quantum signal propagation, and QUBO/QAOA-based optimization were integrated into a unified hybrid AI-quantum framework.

The hybrid framework substantially outperformed every individual method family, achieving a mean top-5 success of approximately 0.952 compared with 0.274 for classical baselines, 0.317 for AI-only models, and 0.413 for quantum-only methods. It also achieved the strongest mean best-hit rank of approximately 2.62, compared with 20.02, 18.95, and 17.54 for classical, AI-only, and quantum-only approaches, respectively. These results are particularly important because practical drug-discovery workflows depend not only on whether validated residues are recovered, but also on how early they appear within ranked candidate lists.

Table 18: Summary of hybrid AI-quantum performance improvements.

Evidence Dimension	Baseline Result	Hybrid Result
Mean top-5 success	0.274 / 0.317 / 0.413	0.952
Mean best-hit rank	20.02 / 18.95 / 17.54	2.62
Search-space reduction	Full graph required	90–99%+
Hard-case recovery	Classical top-5 = 0.0	Hybrid top-5 = 1.0
Seed robustness	Higher variance	1.000 ± 0.000
Scaling evidence	State-vector / proxy scaling	28 qubits, 6.96× speedup

A central scientific insight is that the quantum layer contributes the greatest value not as a standalone replacement for classical or AI methods, but as a targeted refinement mechanism for non-local biological communication. Quantum graph propagation produced residue rankings that differed substantially from classical diffusion-based rankings, indicating that the framework captures distinct long-range communication behavior across protein residue networks. This is biologically meaningful because allosteric regulation frequently depends on distal signaling pathways rather than immediate active-site proximity alone. By modeling the residue graph as a graph-Laplacian Hamiltonian system, the framework captures non-local amplitude propagation behavior beyond classical local averaging processes. QUBO/QAOA optimization further extends this capability by transforming residue discovery into coordinated subset optimization instead of independent residue ranking.

The ablation, hard-case, and robustness analyses strongly support the importance of combining graph structure, AI prioritization, and quantum refinement. Although several ablation variants achieved similar top-5 recovery on reduced evaluation subsets, the full AI-quantum-graph configuration consistently achieved substantially stronger ranking-sensitive metrics. Hard-case cardiac myosin scenarios further demonstrated the practical importance of the hybrid formulation: classical baselines failed under edge-dropout, low-signal, and reduced-information settings, while the hybrid framework maintained successful recovery across

all evaluated perturbation conditions. Statistical evaluation also confirmed strong separation between the hybrid framework and competing baselines, with the hybrid method achieving a mean top-5 success of approximately 0.952 and a 95% confidence interval of [0.857, 1.000].

The corrected biological validation analysis addresses a key limitation of purely computational benchmarking. Across the three strict validation targets, the hybrid ranking recovered known regulatory residue numbers within the top-5 predictions, with the first validation-backed residue appearing at rank 1 for KRAS, BCR-ABL1, and cardiac myosin. This strengthens the claim that the framework is biologically meaningful, not only computationally performant. c-Myc remains appropriately treated as an exploratory qualitative target, preserving a clear distinction between validation-backed recovery and hypothesis generation.

The Fujitsu scaling experiments further demonstrated that the workflow was designed with scalability and distributed simulation in mind. State-vector scaling experiments reached approximately 28 qubits, while MPI rank scaling achieved runtime reductions approaching $6.96\times$ under distributed execution. Communication-aware FusedSWAP-style optimization further reduced runtime overhead by approximately 25–37% across evaluated settings. These scaling results should be interpreted alongside separately verified multi-cpu mpiQulacs logs, which provide the primary evidence for confirmed Fujitsu QARP execution. Together, these results support the feasibility of scalable hybrid AI-quantum workflows for non-local biological discovery problems and establish a realistic pathway toward future quantum-enhanced computational biology systems.

Several limitations remain. Strict quantitative validation was performed primarily on KRAS, BCR-ABL1, and cardiac myosin, while c-Myc was retained as an exploratory qualitative target. In addition, the current framework relies on classical state-vector simulation rather than physical quantum hardware and primarily models static residue-contact graphs. Future work should therefore incorporate molecular dynamics, conformational ensembles, experimentally derived communication pathways, deeper QAOA circuits, and larger residue candidate spaces.

Overall, the findings support a clear conclusion: hybrid AI-quantum modeling is strongest when each computational layer contributes according to its natural strengths. Classical graph analysis provides interpretable structural priors, AI provides search-space reduction and learned prioritization, quantum propagation contributes non-local communication modeling, and QUBO/QAOA optimization enables coordinated residue-subset discovery. Together, these components form a scalable, biologically meaningful, and evidence-backed framework for next-generation allosteric discovery and quantum-enhanced biological search.

Importantly, the current study should be interpreted as a hybrid computational discovery benchmark rather than a definitive biological validation study. The framework demonstrates strong evidence that AI-guided quantum refinement can improve non-local residue prioritization within the evaluated benchmark set, while broader experimental and large-scale biological validation remain future work.

7. Fujitsu QARP Utilization and Distributed Quantum Execution

A major objective of this project was not only to apply quantum-inspired biological modeling, but to demonstrate a realistic hybrid AI-quantum workflow using the Fujitsu Quantum Simulator ecosystem and mpiQulacs-oriented execution infrastructure. The framework integrates quantum graph propagation, AI-weighted QUBO construction, QAOA-based residue selection, MPI-rank scaling analysis, and communication-aware circuit optimization into a unified high-performance computational pipeline.

Unlike small proof-of-concept quantum biology demonstrations, the system was evaluated on large

protein-scale residue graphs containing up to 2,528 residues and 11,439 weighted graph edges. The quantum workflow follows the sequence:

Protein graph $\rightarrow$ Graph-Laplacian Hamiltonian $\rightarrow$ Quantum propagation $\rightarrow$ AI-weighted QUBO $\rightarrow$ QAOA optimization

Quantum components were evaluated using state-vector simulation, MPI-rank scaling analysis, and communication-aware optimization. Separately verified multi-cpu mpiQulacs logs should be treated as the primary evidence for confirmed Fujitsu distributed execution, while the generated scaling plots provide state-vector and proxy-scaling evidence.

State-vector scaling experiments reached approximately 28 qubits, with runtime increasing from approximately 0.034 seconds at 18 qubits to approximately 35.42 seconds at 28 qubits. Memory usage increased from approximately 96 MB to approximately 4.2 GB, consistent with the expected exponential scaling behavior of state-vector simulation.

Table 19: State-vector scaling experiments across increasing qubit counts.

Qubits	Runtime (s)	Memory (MB)	Simulation Mode
18	0.034	96	State-vector scaling
20	0.121	104	State-vector scaling
22	0.612	144	State-vector scaling
24	2.341	352	State-vector scaling
26	8.791	1110	State-vector scaling
28	35.420	4192	State-vector scaling

MPI rank-scaling experiments demonstrated substantial runtime reduction as parallel execution ranks increased. Runtime decreased from approximately 0.879 seconds at one MPI rank to approximately 0.126 seconds at eight ranks, corresponding to an observed speedup of approximately $6.96\times$.

Table 20: MPI rank-scaling performance across distributed execution settings.

MPI Ranks	Runtime (s)	Speedup
1	0.879	1.00 $\times$
2	0.462	1.90 $\times$
4	0.243	3.62 $\times$
8	0.126	6.96 $\times$

Communication-aware optimization was evaluated using FusedSWAP-style circuit optimization strategies designed to reduce global-local qubit communication overhead during distributed-style simulation. Across evaluated configurations, this reduced runtime by approximately 25–37%, demonstrating the importance of qubit-layout-aware execution for scalable hybrid quantum workloads.

Table 21: Communication-aware FusedSWAP-style optimization results.

Qubit Setting	Runtime Reduction	Optimization
10 qubits	36.7%	FusedSWAP-style
15 qubits	29.8%	FusedSWAP-style
20 qubits	25.1%	FusedSWAP-style
22 qubits	25.0%	FusedSWAP-style
24 qubits	25.0%	FusedSWAP-style

The framework also generated end-to-end quantum evidence artifacts across all benchmark proteins, including graph-Laplacian propagation outputs, AI-weighted QUBO objective files, QAOA residue-subset optimization outputs, hybrid quantum scoring tables, and scaling evidence.

Table 22: Quantum module evidence generated across the hybrid workflow.

Quantum Component	Evidence Generated
Graph-Laplacian Hamiltonian	Quantum propagation amplitude files
Quantum propagation	Non-local residue communication scores
AI-weighted QUBO	Structured optimization objective files
QAOA optimization	Residue subset sampling outputs
Hybrid quantum ensemble	Quantum-enhanced residue rankings
mpiQulacs / QARP execution	Separately verified multi-cpu execution logs
FusedSWAP-style optimization	Communication-aware runtime reduction

Importantly, this work does not claim quantum supremacy, fault-tolerant quantum advantage, or physical-hardware quantum advantage. Instead, it demonstrates a realistic hybrid AI-quantum computational workflow in which the quantum-method layer acts as a structured refinement mechanism for non-local biological discovery. The significance of the framework lies in the integration of biological graph intelligence, machine learning candidate prioritization, quantum propagation dynamics, combinatorial optimization, and Fujitsu-oriented scaling analysis into a single reproducible discovery engine.

Taken together, these results establish meaningful Fujitsu QARP utilization through distributed-execution design, MPI-aware scaling analysis, communication optimization, and scalable hybrid AI-quantum biological computation rather than isolated simulator invocation alone.

8. Business Impact and Applications

The proposed AI-quantum allosteric discovery framework addresses a high-value bottleneck in drug discovery: identifying distal regulatory residues that can modulate protein function without directly targeting the active site. This is commercially important because many disease-relevant proteins are difficult to inhibit through conventional active-site binding due to poor selectivity, resistance mutations, shallow binding pockets, or competition with native substrates. Allosteric targeting offers an alternative route by identifying regulatory regions that can tune protein activity more selectively.

In practice, this positions the framework as a decision-support layer before costly mutagenesis campaigns, molecular-dynamics simulations, docking workflows, and experimental validation studies.

The framework provides business value in three main ways: reducing experimental search cost, improving early-hit prioritization, and expanding the set of druggable hypotheses for difficult targets. Across the benchmark, AI filtering reduced candidate residue search spaces by more than 90–99%, including 99.39% for BCR–ABL1, 98.53% for KRAS, and 99.80% for cardiac myosin in top-5 candidate mode. Instead of experimentally or computationally screening thousands of residues, discovery teams can therefore focus on compact high-confidence candidate regions.

The hybrid framework also improved early-hit quality. It achieved a mean top-5 success of approximately 0.952, compared with 0.274 for classical graph baselines, 0.317 for AI-only models, and 0.413 for quantum-only methods. This is directly relevant to pharmaceutical workflows because early ranked candidates drive decisions involving mutagenesis, structural validation, pocket mapping, ligand screening, and medicinal chemistry investment. Stronger early-hit ranking can substantially reduce wasted validation cycles and accelerate movement from structural hypothesis to experimentally testable allosteric sites.

A major application is allosteric drug-target discovery. For targets such as KRAS, BCR–ABL1, cardiac myosin, and exploratory c-Myc, the framework generates ranked candidate residues that may define regulatory pockets or communication hotspots. These outputs can guide site-directed mutagenesis, fragment screening, molecular docking, cryo-EM interpretation, and follow-up molecular dynamics studies.quantum-inspired

Table 23: Business relevance of the hybrid AI–quantum framework.

Business sion	Dimen-	Project	Result	Practical Value
Candidate reduction		90–99%+	search-space compression	Lower screening and validation burden
Early-hit recovery		0.952	hybrid top-5 success	Improved prioritization of experimental targets
Ranking quality		Best-hit	rank improved to 2.62	Validated residues appear earlier in candidate lists
Hard-case recovery		Classical 1.0	0.0, hybrid	Supports difficult distal/allosteric cases
Scalability evidence		28-qubit	state-vector scaling, 6.96× MPI-rank speedup	Supports future high-complexity workloads
Communication optimization		25–37%	runtime reduction	Improves feasibility of larger simulations

A second application is early-stage target validation. Pharmaceutical discovery teams frequently need to decide which residues or regions justify deeper experimental investment. The proposed system provides not only a ranked candidate list, but also multiple evidence streams including graph importance, AI confidence, quantum propagation signal, QUBO/QAOA selection behavior, perturbation robustness, and final hybrid ranking scores. This makes the output substantially more actionable than isolated black-box prediction models.

A third application is drug-resistance and pathway modulation analysis. Because the framework models non-local communication across protein residue graphs, it can help identify residue networks associated with conformational switching, pathway signaling, and resistance-linked structural behavior. This is especially relevant for oncology targets, kinases, GTPases, and proteins where active-site inhibition alone is insufficient

or resistance-prone.

The framework additionally has reusable platform value beyond a single protein system. The same computational pipeline can be applied to new proteins by converting structures into residue graphs, applying AI candidate filtering, executing quantum propagation, and performing QUBO/QAOA residue selection. This makes the framework suitable as a reusable discovery engine for pharmaceutical companies, biotechnology teams, and computational biology research groups.

From an operational perspective, the framework integrates naturally into existing computational drug-discovery workflows. It can be deployed before expensive wet-lab validation, before large molecular-dynamics campaigns, or before computationally intensive docking studies. By narrowing candidate regions early, the framework can reduce downstream experimental burden and improve allocation of computational and laboratory resources.

The Fujitsu QARP component further strengthens the practical relevance of the system by demonstrating scalable hybrid AI-quantum workflow design. State-vector/proxy scaling up to approximately 28 qubits, MPI-rank scaling approaching $6.96\times$ speedup, and communication-aware runtime reductions of approximately 25–37% collectively support the feasibility of future large-scale hybrid biological discovery pipelines. Importantly, the framework does not claim fault-tolerant quantum advantage or physical quantum supremacy. Instead, it demonstrates a realistic quantum-assisted computational workflow capable of evolving alongside future HPC and quantum infrastructure.

Overall, the business impact of this work lies in the creation of a scalable AI-quantum decision-support framework for allosteric discovery. The system reduces candidate-search burden, improves early-hit recovery, supports difficult target exploration, and provides interpretable evidence for experimental prioritization. These capabilities make the framework relevant to pharmaceutical R&D, biotechnology discovery platforms, translational computational biology, and future quantum-enhanced scientific computing pipelines.

9. Conclusion and Future Work

This work presented a scalable hybrid AI-quantum framework for allosteric residue discovery using graph learning, machine learning, quantum propagation, and QUBO/QAOA-based optimization. Across biologically diverse protein systems, the framework converted protein structures into residue interaction graphs, reduced candidate search complexity through AI prioritization, modeled non-local communication using graph-Laplacian quantum propagation, and refined residue selection through structured quantum optimization.

The results demonstrate that the strongest performance emerges from hybrid integration rather than from any individual method family alone. Classical graph baselines provided interpretable structural priors, AI reduced the search space by more than 90–99%, quantum propagation introduced distinct non-local residue communication signals, and QUBO/QAOA optimization enabled coordinated residue-subset selection. Together, these layers achieved a mean top-5 success of approximately 0.952, substantially outperforming classical (0.274), AI-only (0.317), and quantum-only (0.413) approaches, while improving the mean best-hit rank to approximately 2.62.

Table 24: Final hybrid AI–quantum framework summary.

Project Dimension		Final Outcome
Protein scale	benchmark	Up to 2528 residues and 11,439 graph edges
AI search-space reduction		Greater than 90–99% candidate compression
Hybrid top-5 success		Approximately 0.952
Hybrid best-hit rank		Approximately 2.62
Quantum scaling evidence		Approximately 28-qubit state-vector/proxy scaling
MPI-rank scaling		Approximately $6.96\times$ MPI-rank speedup
Communication optimization		Approximately 25–37% runtime reduction

A key scientific insight is that quantum methods contribute most effectively as targeted refinement layers for non-local and combinatorial biological discovery problems rather than as standalone replacements for classical computation. In this framework, AI compresses the search space, quantum propagation captures long-range communication behavior, and hybrid evidence fusion produces biologically interpretable residue rankings.

The Fujitsu QARP and mpiQulacs-oriented scaling experiments further demonstrated scalable workflow design. State-vector/proxy scaling reached approximately 28 qubits, MPI-rank scaling achieved speedups approaching $6.96\times$, and FusedSWAP-style optimization reduced runtime by approximately 25–37%. Generated scaling plots provide state-vector/proxy scaling evidence, while separately verified `multi-cpu mpiQulacs` logs serve as the primary evidence for confirmed Fujitsu QARP execution.

9.1. Future Work

Future work should extend the framework in the following directions:

- **Expand biological validation.** Future studies should extend benchmarking beyond KRAS, BCR–ABL1, and cardiac myosin by incorporating additional protein families and stronger experimentally validated allosteric annotations.
- **Move from static graphs to dynamic ensembles.** The current framework uses static residue-contact graphs. Future versions should integrate molecular dynamics, conformational sampling, cryo-EM states, AlphaFold ensemble variation, and experimentally derived communication pathways.
- **Deepen the quantum layer.** Future experiments should evaluate larger residue candidate spaces, deeper QAOA circuits, alternative Hamiltonian encodings, improved QUBO penalty tuning, and more rigorous sampling analysis.
- **Strengthen Fujitsu QARP execution evidence.** Future work should include larger distributed workloads with verified `multi-cpu mpiQulacs` execution logs, higher MPI-rank scaling, and direct comparisons between optimized and unoptimized circuit configurations.

- **Connect predictions to downstream drug-discovery workflows.** Predicted residues should be integrated into mutagenesis planning, docking, fragment screening, molecular-dynamics follow-up, and wet-lab validation workflows to strengthen practical pharmaceutical relevance.

Overall, this work establishes a technically grounded pathway toward future quantum-enhanced biological discovery. The framework reduces experimental search burden, improves early-hit recovery, captures difficult distal communication signals, and produces multi-evidence residue rankings suitable for next-generation allosteric drug-discovery pipelines. The final contribution is therefore not merely a quantum biology demonstration, but a scalable hybrid AI–quantum computational discovery engine for non-local biological search.

10. References

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